Momentum

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Calculate and compare signed momentum.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 15.*
Need orbital geometry/positions and phase or time reference. A period alone does not show simultaneous nearby positions; uncertainties matter.

02. Worked example: explain the model
Both left; velocity 2 m/s, momentum 6 kg·m/s.

03. Read and interpret the model
Rightward cart positive; leftward cart negative.

04. Apply the physics
+6 and -6; total 0 kg·m/s. Both are moving.

05. Explain your reasoning
Both double momentum magnitude if direction unchanged.

06. Review: connect an earlier idea *Review: Unit 2, Day 6.*
g=6 N/kg; force 48 N toward the source.

07. Exit check
8 kg·m/s left.

08. STRETCH: test the model
Example 1 kg at +6 m/s and 3 kg at -2 m/s; momenta +6 and -6.

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01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 15.*
Overlap days 15-17. Timing is compatible, but spatial geometry, miss distance and uncertainties are still needed.

02. Worked example: explain the model
Both left; velocity 2 m/s, momentum 6 kg·m/s.

03. Read and interpret the model
Rightward cart positive; leftward cart negative.

04. Apply the physics
+3 and -3 kg·m/s; total zero. Equal opposite momentum arrows cancel, although both carts move.

05. Explain your reasoning
Both double momentum magnitude if direction unchanged.

06. Review: connect an earlier idea *Review: Unit 2, Day 6.*
Field 4 N/kg; force 3.2 N toward the source.

07. Exit check
v=-4 m/s; new p=+12 kg·m/s.

08. STRETCH: test the model
4-10+3 v=0 gives v=+2 m/s. After reversing first cart, total -4-10+6=-8 kg·m/s.